

AnySat: One Earth Observation Model for Many Resolutions, Scales, and Modalities

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Abstract

Geospatial models must adapt to the diversity of Earth observation data in terms of resolutions, scales, and modalities. However, existing approaches expect fixed input configurations, which limits their practical applicability. We propose AnySat, a multimodal model based on joint embedding predictive architecture (JEPA) and scale-adaptive spatial encoders, allowing us to train a single model on highly heterogeneous data in a self-supervised manner. To demonstrate the advantages of this unified approach, we compile GeoPlex, a collection of 5 multimodal datasets with varying characteristics and 11 distinct sensors. We then train a single powerful model on these diverse datasets simultaneously. Once fine-tuned or probed, we reach state-of-the-art results on the test sets of GeoPlex and for 6 external datasets across various environment monitoring tasks: land cover mapping, tree species identification, crop type classification, change detection, climate type classification, and segmentation of flood, burn scar, and deforestation. Our code and models are available at <https://github.com/gastruc/AnySat>.

1. Introduction

From a remote sensing perspective, the natural images of computer vision are remarkably uniform: they are captured by nearly identical sensors (standard cameras) with the same RGB channels and are often taken from similar perspectives. This consistency allows the creation of large composite image datasets from various sources [19, 39, 46], which are key for image foundation models to learn powerful, general-purpose features [5].

In contrast, Earth observation (EO) data displays significant variability in modalities, scales, and spatial, temporal, and spectral resolutions. Existing EO foundation models are generally trained on a single dataset with a specific format [8, 25, 33, 52], and cannot be applied to datasets with different input types without retraining from scratch—defeating

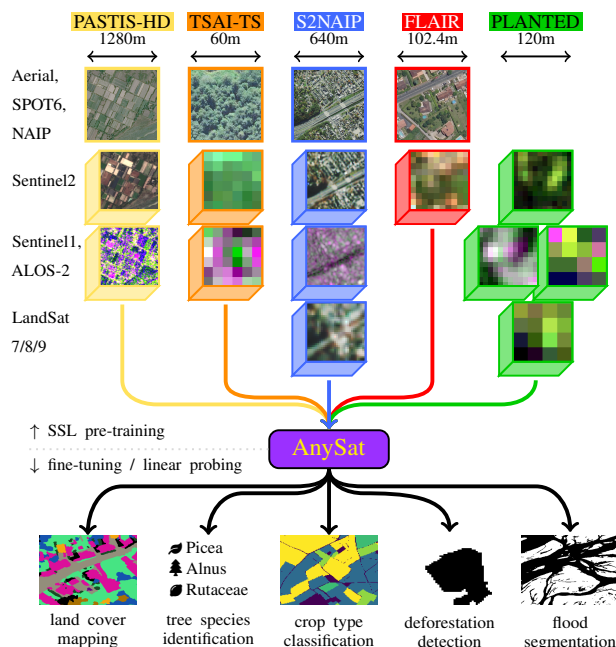


Figure 1. **Multi-Dataset Training.** For the first time, a single model can be pretrained **simultaneously** on a collection of Earth Observation datasets with heterogeneous resolutions, scales, and modalities. The resulting model can be fine-tuned to achieve state-of-the-art results for a wide variety of data types and tasks.

the purpose of foundation models. EO foundation models should be able to seamlessly integrate new datasets for training and prediction, regardless of their resolution, scale, and modalities. As recent efforts provide more flexibility in terms of modalities [4, 29], scale [42], or spectral resolutions [54], none fully leverage the diversity of EO sensors.

We introduce **AnySat**, a novel EO model using the spatial alignment of multiple modalities as a source of self-supervision. Indeed, while multiple observations of the same area from distinct sensors capture different information, they share the same underlying semantics. Therefore, we can expect the learned representations to be consistent across

modalities. Moreover, we should be able to reconstruct missing modalities from available ones, encouraging the use of cross-modal masked auto-encoding techniques [4, 27]. However, EO data are subject to complex disruptors such as weather conditions, acquisition angles, and variations in time of day or year. To overcome this issue, we design a new multimodal Joint Embedding Predictive Architecture (JEPA) [3] to learn representations that are consistent *in feature space*.

A key advantage of our JEPA model is that it eliminates the need for modality-specific decoders, allowing us to handle a wide variety of sensors seamlessly. Combined with our scale-adaptive patch encoder architecture, this approach enables us to train a single model on highly heterogeneous collections of multimodal EO datasets. Notably, over 75% of the learnable parameters in our model are shared across all modalities and resolutions, and thus fully benefit from large and varied training data for self-supervision.

To evaluate our approach, we compile **GeoPlex**, a collection of 5 multimodal datasets including 11 distinct modalities, with aerial images and satellite time series, radar and optical sensors. GeoPlex spans various spatial resolutions (from 0.2 to 250 m per pixel), revisit times (from single images to weekly time series), channel counts (3 to 11), and spatial extent (samples ranging from 0.4 to 160K hectares). To showcase the versatility of AnySat, we also consider 6 external evaluation datasets with diverse characteristics. After fine-tuning, AnySat achieves state-of-the-art performance on 9 downstream tasks, including classification, segmentation, and change segmentation across domains such as land cover mapping, crop type classification, tree species identification, and deforestation detection. Our contributions are as follows:

- We present AnySat, a versatile architecture capable of learning from multiple EO sources with heterogeneous resolutions, scales, and modalities.
- We introduce the first application of JEPA for multimodal EO data, enabling large-scale and efficient self-supervised learning.
- We demonstrate that, when pretrained on a curated collection of EO datasets, AnySat can be fine-tuned or linearly-probed to achieve state-of-the-art performance across a diverse array of tasks and datasets.

Thanks to its flexible design, our pretrained model can be applied to scales ranging from a single forest plot to tiles covering hundreds of square kilometers, and adapt to diverse sensor setups—from unimodal data to any combination of the 11 sensors featured in GeoPlex. In addition, we demonstrate that AnySat successfully generalizes to new sensor configurations not present in its training set.

2. Related Work

In this section, we review the dynamic field of self-supervised learning in geospatial models, highlighting recent

efforts to enhance their adaptability to diverse inputs. Finally, we present the feature-predictive paradigm, which is instrumental to improve the versatility of EO models.

Self-Supervised Geospatial Models. The abundance of raw EO data makes it particularly suitable for self-supervised learning approaches [6, 10, 34, 50]. Generative models leverage the unique properties of EO data with adapted strategies such as spectral [14], temporal [16, 17], and spatio-temporal [29, 56], and hybrid [49] masking. Other approaches predict rotated [32] or rescaled [36, 42, 48] versions of the input data, or predict missing modalities from available ones [4, 18]. However, these models are often trained on specific combinations of modalities and are limited to those modalities during inference, which hinders their applicability as foundation models expected to adapt to diverse scenarios.

Versatile EO Models. Several approaches have been proposed to improve the generalizability of EO models. Some models address variability in spatial resolutions by training on images of different resolutions and generalizing to coarser scales [42], while others manage spectral variability by training on sensors with different spectral bands [54]. Temporal adaptability is achieved in models capable of handling both single-date images and image time series [4, 8, 25]. Attempts have also been made to generalize across modalities by training on data from different sensors [28, 29] or and even text or audio [45]. Despite these efforts, many models are still trained with a single scale and expect the input to have a certain shape, typically 224×224 pixels. They resize other inputs to fit the model architecture, leading to inefficiencies for smaller inputs [49, Tab 5]. A key obstacle preventing the creation of truly versatile generative self-supervised models is the requirement for multiple encoders, decoders, and augmentations to handle different configurations. In this paper, we explore feature-predictive architectures as a promising solution to this challenge.

Feature-Predictive Architectures. Self-supervised learning methods have achieved significant success in image analysis [12, 26, 39]. These approaches learn without labels using pretext tasks, which can be discriminative [23, 37], contrastive [11, 12, 24, 26], or generative, where the model predicts a degraded version of its input [27, 51]. Recent works have proposed performing reconstruction in feature space rather than input space (e.g., pixel space) [7, 55]. Among feature-predictive architectures, the Joint Embedding Predictive Architecture (JEPA) has shown particular promise [3] by learning to predict the features of masked parts of an input image. Feature space reconstruction based model can also be combined with contrastive objectives for improved stability and representation quality [7].

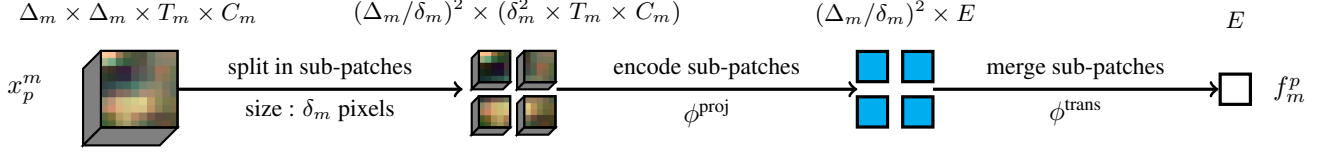


Figure 2. **Scale-Adaptive Patch Encoding.** We consider a patch x_p^m of resolution $\Delta_m = P/R_m$ pixels. We first split x_p^m into sub-patches of size δ_m pixels, which are mapped by a modality-specific projector ϕ_m^{proj} to a E -dimensional embedding. Then, a shared spatial transformer module ϕ^{trans} combines all sub-patches into a vector of size E . As the sub-patch size δ_m is fixed, the patch sizes Δ_m only influences the number of input tokens to ϕ^{trans} , allowing us to use the same network for different resolutions.

Because it bypasses the need for complex data augmentations or decoder networks, JEPA is particularly well-suited for massively multimodal applications such as Earth observation. SAR-JEPA [31] introduces the first implementation of JEPA concepts for EO, focusing exclusively on SAR data. In this paper, we combine JEPA with a versatile spatial encoder architecture, allowing a single model to handle diverse data scales, resolutions, and modalities.

3. Method

We first describe our proposed architecture (Sec. 3.1) and self-supervised training procedure (Sec. 3.2). Then, we detail the fine-tuning and probing methods used for downstream tasks (Sec. 3.3). Our work focuses primarily on multi-dataset self-supervised training. However, for clarity, we initially describe the method for a single multimodal dataset, later generalizing it to multiple datasets.

3.1. Architecture

Tiles with multimodal observations are first partitioned into spatially aligned patches. Unlike classical Vision Transformers [15], our model supports patches of varying sizes, accommodating the significant scale variations common in Earth Observation (EO) datasets. Each patch is embedded via a scale-adaptive patch encoder, after which a combiner network integrates representations from multiple modalities into a unified spatial embedding.

Formally, we consider a tile x of size $S \times S$ meters, observed through multiple modalities $\mathbf{M}$. Each modality $m \in \mathbf{M}$ has its own resolution R_m (meters per pixel), temporal observations T_m (with $T_m = 1$ for single-date modalities), and number of channels C_m (e.g., spectral or polarization channels). The tile x observed in modality m is denoted x^m and is represented as a tensor of shape $(S/R_m) \times (S/R_m) \times T_m \times C_m$.

Spatially Consistent Patching. Tiles are partitioned into a set $\mathbf{P}$ of non-overlapping patches, each of size $P \times P$ meters. An input token x_p^m represents the observation of patch $p \in \mathbf{P}$ in modality m . All modalities share the same spatial patch layout, ensuring spatial consistency across modalities. The

total number of tokens is thus $|\mathbf{M}| \cdot (S/P)^2$. Although the patch size is constant across modalities, each token may have distinct tensor dimensions due to differing resolutions, temporal extents, and channel numbers.

Patch Encoding. We design a scale-adaptive patch encoder ϕ^{patch} to map each input token x_p^m into a fixed-size vector $f_p^m \in \mathbb{R}^E$, regardless of modality resolution R_m or patch size P . The encoding scheme, illustrated in Fig. 2, involves three stages:

- (i) We first subdivide each token into fixed-size sub-patches of $\delta_m \times \delta_m$ pixels, flattening their spatial dimensions to vectors of size $\delta_m^2 T_m C_m$.
- (ii) Each flattened sub-patch is mapped to dimension E via a modality-specific MLP ϕ_m^{proj} . For multi-temporal modalities ($T_m > 1$), a Lightweight Temporal Attention Encoder (LTAE)[21] collapses the temporal dimension.
- (iii) We add positional encodings based on ground sampling distance [42] to the sub-patch embeddings. A shared transformer network ϕ^{trans} with B blocks aggregates the sub-patch embeddings into a single representation f_p^m per modality using a CLS-like token.

Using sub-patches of fixed sizes δ_m allows ϕ^{patch} to process patches of different patch sizes P without rescaling the input data. Indeed, changes in P only influence the number of input tokens processed by ϕ^{trans} , which has no incidence on the embedding size.

Modality-Combiner Network. The combiner network ϕ^{comb} merges embeddings f_p^m from all available modalities into a multimodal representation f_p^* for each patch $p \in \mathbf{P}$. We use the cross-attention-based architecture proposed by OmniSat [4, 3.1]: (i) We first add to each f_p^m an absolute positional encoding $\text{pos}(p)$ —the same one used for sub-patches.; (ii) The tokens go through a sequence of B self-attention blocks; (iii) We associate each patch with a token with a shared learned value and add positional encoding; and (iv) We compute the cross-attention between these tokens and the embeddings of the last self-attention block. This results in one embedding per patch f_p^* .

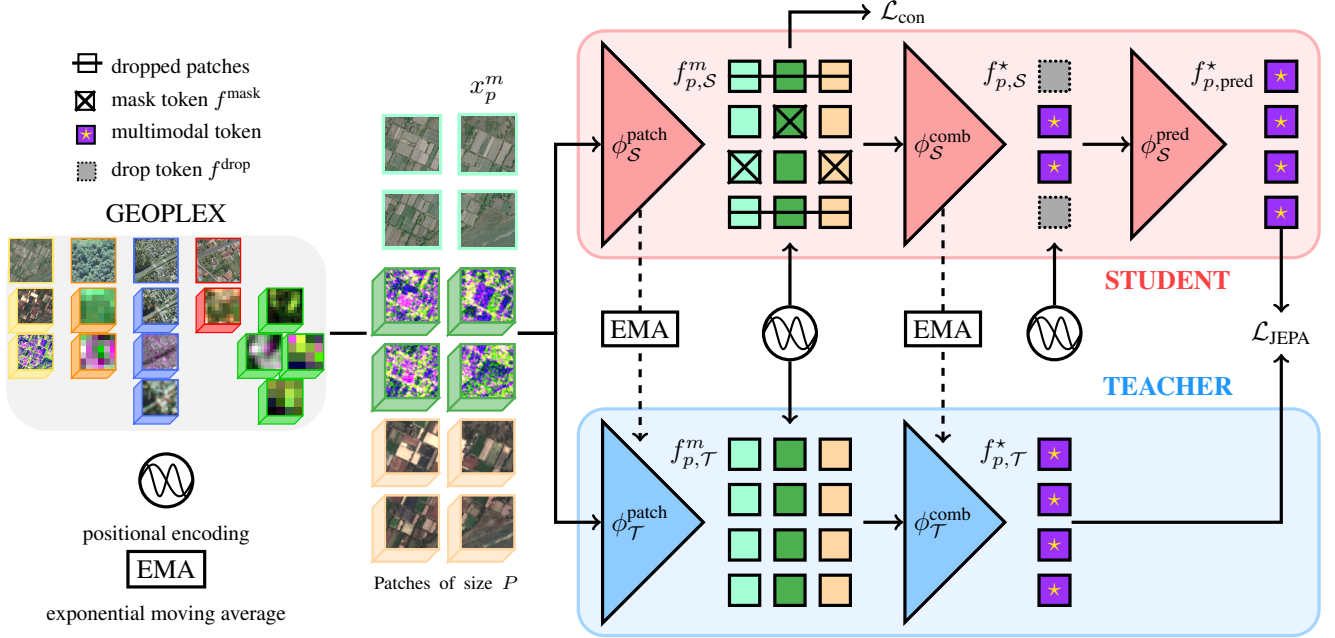


Figure 3. **Architecture of AnySat.** We begin each iteration by randomly selecting a dataset among GeoPlex and sampling a tile. Each available modality is divided into spatially aligned patches of size P . The student network’s patch encoder ϕ_S^{patch} embeds each patch and we apply a contrastive loss to encourage spatial consistency across modalities. We then apply dropping and masking : some patches have all modalities removed (dropping), while others have only random modalities removed (masking). The remaining patches are merged in the modality combiner ϕ_S^{comb} to form multimodal representations $f_{p,S}^*$ for the non-dropped patches. The predictor ϕ_S^{pred} then reconstructs the embeddings of the dropped patches. Finally, the student network’s output is compared to the teacher’s, whose weights are an Exponential Moving Average (EMA) of the student’s weights and which processes the complete set of patches without masking or dropping.

3.2. Training

We adapt the Joint Embedding Predictive Architecture (JEPA) framework [3] to multimodal Earth Observation, enabling self-supervised pretraining on datasets of varying modalities without labels. A student network operates on heavily masked inputs, aiming to predict embeddings generated by an unmasked teacher network whose parameters follow an Exponential Moving Average (EMA) of the student’s weights [26]. Training leverages two losses: a contrastive loss to enforce modality consistency and a JEPA loss for masked embedding prediction.

The student network consists of a patch encoder ϕ_S^{patch} , a modality combiner ϕ_S^{comb} , and a predictor network ϕ_S^{pred} with 3 self attention blocks. The teacher network includes a patch encoder ϕ_T^{patch} and a modality combiner ϕ_T^{comb} and no predictor. The student network first embeds all input tokens x_p^m into vectors of size E using the patch encoder:

$$f_{p,S}^m = \phi_S^{\text{patch}}(x_p^m). \quad (1)$$

Contrastive Loss. For a fixed patch p , the observations x_p^m for $m \in \mathcal{M}$ capture different aspects of the same spatial region but share the same underlying semantics: the content

of p . Therefore, we expect the representations $f_{p,S}^m$ to be consistent across modalities. We enforce this intuition with a contrastive loss inspired by OmniSat [4]. Specifically, we use a modified InfoNCE loss [38], where each token (p, m) is positively paired with those from the same spatial patch but different modalities:

$$\mathcal{L}_{\text{con}} = \sum_{(p,m) \in \mathbf{P} \times \mathbf{M}} \frac{-\log}{|\mathbf{P}||\mathbf{M}|} \left(\frac{\sum_{n \neq m} \exp(\langle f_{p,S}^m, f_{p,S}^n \rangle / \tau)}{\sum_{\substack{n \neq m \\ q \neq p}} \exp(\langle f_{p,S}^m, f_{q,S}^n \rangle / \tau)} \right), \quad (2)$$

where τ is a temperature parameter, and $\langle \cdot, \cdot \rangle$ denotes the cosine similarity between embeddings.

Joint Embedding Predictive Architecture. We adapt the JEPA self-supervised learning framework [3] to the context of multimodal Earth Observation. Avoiding reconstruction in pixel space is particularly beneficial for EO data, which can be heavily influenced by factors such as weather, time of day, or acquisition angle. Reconstructing in latent space allows us to learn more consistent and semantically meaningful features. The training process proceeds as follows:

- **Patch Dropping.** We apply JEPA’s masking strategy by randomly selecting five rectangular regions on the tile. Let $\mathbf{K} \subset \mathbf{P}$ be the set of patches intersected by these rectangles, and $\bar{\mathbf{K}} = \mathbf{P} \setminus \mathbf{K}$ the remaining patches. We drop all the student’s tokens $f_{p,S}^m$ for patches $p \in \mathbf{K}$.
- **Modality & Temporal Masking:** We randomly mask a subset $\mathbf{L} \subset \bar{\mathbf{K}} \times \mathbf{M}$ of the remaining tokens, ensuring that at least one modality per patch remains unmasked. Masked token embeddings are replaced with a fixed value $f^{\text{mask}} \in \mathbb{R}^E$, which is learned as a parameter of the network. We also randomly mask 50% of the timestamps of all time series.
- **Combiner:** We input all tokens (masked or not) to the student’s combiner $\phi_{\mathcal{S}}^{\text{comb}}$, producing multimodal embeddings $f_{p,S}^*$ for all $p \in \bar{\mathbf{K}}$:

$$f_{p,S}^* = \phi^{\text{comb}}(\{f_{p,S}^m\}_{(p,m) \notin \mathbf{L}} \cup \{f^{\text{mask}}\}_{(p,m) \in \mathbf{L}}). \quad (3)$$

- **Predictor:** We replace each dropped patch $p \in \mathbf{K}$ with a fixed value $f^{\text{drop}} \in \mathbb{R}^E$. We add positional encodings to all tokens (including the dropped ones) and input them to the predictor ϕ_S^{pred} , yielding embeddings $f_{p,\text{pred}}^*$ for all patches $p \in \mathbf{P}$:

$$f_{p,\text{pred}}^* = \phi_S^{\text{pred}}(\{f_{p,S}^*\}_{p \in \bar{\mathbf{K}}} \cup \{f^{\text{drop}}\}_{p \in \mathbf{K}}). \quad (4)$$

- **Teacher Encoding:** The teacher network receives all input tokens x_p^m , embeds them using $\phi_{\mathcal{T}}^{\text{patch}}$, and combines them with $\phi_{\mathcal{T}}^{\text{comb}}$ without any dropping, masking, or temporal dropout. The teacher outputs patch embeddings $f_{p,\mathcal{T}}^*$ for all $p \in \mathbf{P}$.
- **Loss Function:** The training objective is the L_2 distance between the student predictions and the teacher’s multimodal embeddings for the dropped patches:

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|\mathbf{K}|} \sum_{p \in \mathbf{K}} \|f_{p,\text{pred}}^* - f_{p,\mathcal{T}}^*\|_2^2. \quad (5)$$

After training, we use the teacher network for downstream tasks and discard the student. Note that all modules are shared across all modalities except for the projection layers ϕ_m^{proj} in the patch encoder ϕ^{patch} .

Training with Multiple Datasets. The flexibility of AnySat enables us to train a single model simultaneously on several datasets of various sizes and scales with the same weights and without rescaling. We consider a set $\mathbf{D}$ of such datasets. Each dataset $d \in \mathbf{D}$ is characterized by the subset $M_d \subset \mathbf{M}$ of its available modalities and S_d the size of its tiles. We also consider a batch size B_d and a set P_d of acceptable patch sizes, which depend on the nature of the data, the available resolution, and the tile size. We use the following procedure:

1. Randomly select a dataset d in $\mathbf{D}$.

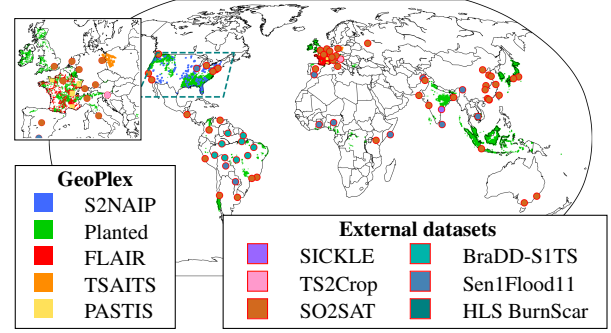


Figure 4. **Datasets Considered.** GeoPlex is composed of 5 diverse dataset spanning the entire world, with a higher concentration in Europe and the US where open-data are more abundant. We also consider external evaluation datasets with a more diverse spread.

2. Randomly select a patch size P in P_d .
3. Randomly sample B_d tiles in d .
4. Process the tiles and backpropagate the loss.

3.3. Downstream Tasks

After pretraining, AnySat can be fine-tuned or probed for various downstream tasks, including classification and semantic segmentation.

Classification. For tile-level classification, we insert a [CLS] token into the combiner network’s cross-attention module. This token generates a tile-level embedding, subsequently mapped to label logits through a linear classifier.

Semantic Segmentation. For semantic segmentation, we predict labels at pixel-level resolution by first selecting a modality whose resolution is close to the annotation resolution. A dense feature map at the sub-patch scale (δ_m) is formed by concatenating sub-patch embeddings (outputs of ϕ_m^{proj}) with corresponding multimodal patch embeddings (outputs of ϕ^{comb}). An MLP then maps these concatenated embeddings to logits of dimension $\delta_m \times \delta_m \times N$, where N is the number of semantic classes. Unfolding these logits yields pixel-level predictions. Using sub-patches results in higher-resolution predictions compared to methods that rely only on patch-level representations.

Probing. AnySat supports linear probing, where a simple linear classifier can be attached directly to the class token for classification or to the dense feature maps for segmentation. This approach avoids complex segmentation heads typically required in earlier methods [35], leveraging the dense features produced by our architecture.

New Sensor Configurations. AnySat can adapt to sensors with configurations differing from those in the training

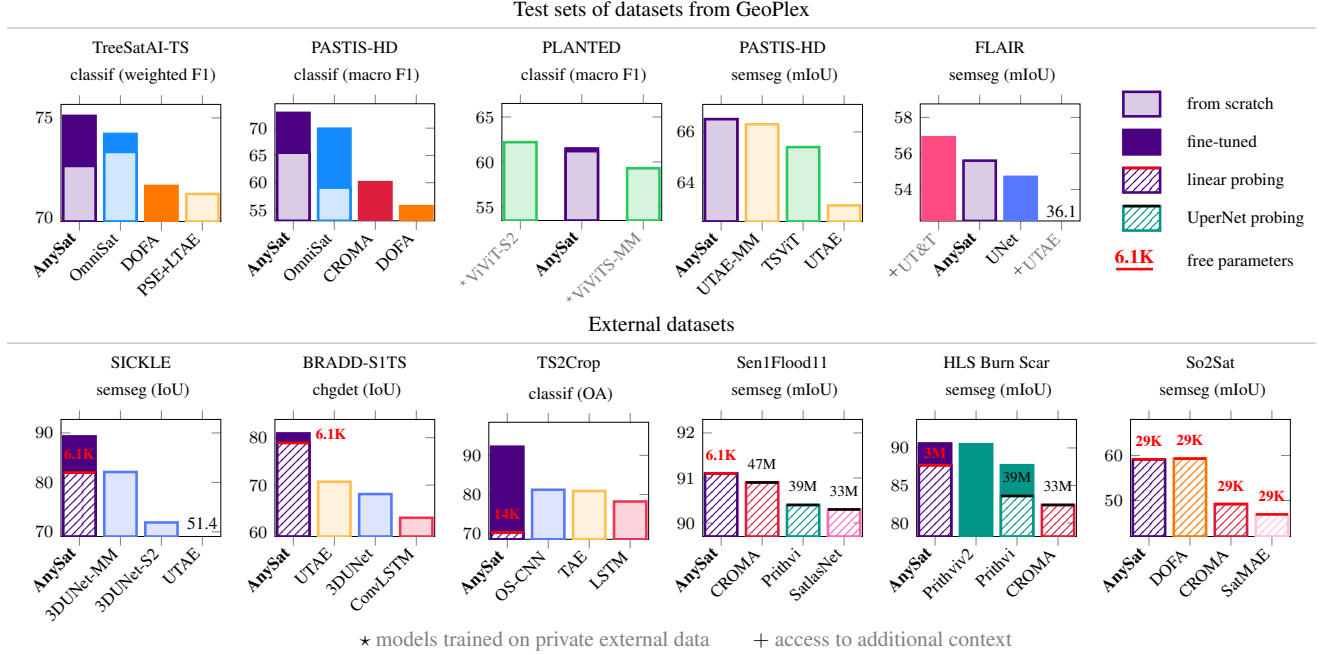


Figure 5. **Quantitative Evaluation.** We evaluate AnySat across 9 open-access datasets and for four tasks: multilabel classification (classif), semantic segmentation (semseg), pixel-wise change detection (chgdet), and pixel-wise regression (regression). For clarity, we only visualize the four best performance per dataset, see Appendix for full results. We report the number of trainable parameters for probing evaluations.

datasets well as new sensors. During self-supervised pre-training, we learn a sensor-specific scalar value representing missing data, which is subsequently used wherever modality channels are absent during fine-tuning or probing. For sensors not featured in the training sensors, we randomly initialize a new projector and fine-tune it along the other free parameters. This effectively extends AnySat to previously unseen sensors but cannot be used in probing for sensors too different from the training mix.

4. Experiments

4.1. Datasets and Evaluation

We present the datasets used for training and evaluation, as well as our evaluation protocol.

GeoPlex. As argued by Roscher *et al.* [43], EO models benefit from high-quality, diverse, and curated data rather than extensive but uniform acquisitions. We follow this principle by compiling a collection of five multimodal datasets, each featuring different combinations of modalities, scales, and resolutions. GeoPlex comprises the training sets of the following datasets:

- **TreeSatAI-TS** [1, 4]: A forest-centric dataset in Germany with Sentinel-1 & 2 time series and Very High Resolution (VHR) images at 0.2 m resolution.
- **FLAIR** [20]: A French land cover dataset with Sentinel-

2 time series and VHR images with elevation data (0.2 m). To form multimodal patches, we crop the Sentinel-2 time series to match the extent of the VHR images (discarding 93.5% of pixels).

- **PLANTED** [40]: A global forest dataset comprising time series from multiple sensors, including Sentinel-1/2, Landsat-7, ALOS-2, and MODIS. Only 1.3 of the 2.3M images used in the paper are publicly available.
- **S2NAIP-URBAN** [8]: An urban dataset in the continental US with VHR images (1.25m) and time series from Sentinel-1/2 and Landsat-8/9.
- **PASTIS-HD** [4, 22]: A French crop mapping dataset with VHR images (1.5m) and Sentinel-1 & 2 time series. As PASTIS is evaluated in 5-fold cross-validation, there are no dedicated train and test sets. We include the entire dataset (without labels) in GeoPlex.

As illustrated in Fig. 4, GeoPlex spans 249K km² across five continents and 171 billion pixels. The sampled tiles range in size from 0.36 to 164 hectares. GeoPlex includes 11 distinct modalities with resolutions ranging from 0.2 m to 250 m, with both VHR images and time series data:

- **Very High Resolution Images:**
 - **Aerial:** RGB+NIR (near-infrared) at 0.2 m
 - **Aerial+NMS:** RGB+NIR+Elevation at 0.2 m
 - **NAIP:** RGB+NIR at 1.25 m
 - **SPOT6:** RGB+NIR at 1.5 m.

- **Time Series Data:**
 - **Sentinel-1:** 3 channels (VV/VH polarization + ratio) at 10 m, Ascending & Descending Orbits
 - **Sentinel-2:** 10 channels at 10 m
 - **ALOS-2:** 3 channels (polarization) at 30 m
 - **Landsat-7:** 6 channels at 30 m
 - **Landsat-8/9:** 11 channels at 30 m
 - **MODIS:** 7 channels at 250 m.

We select the possible patch size per dataset, while we set the sub-patch size per modality 1 pixel for very high-resolution images and 10 pixels for time series data. See the Appendix for the complete characteristics of all datasets.

External Datasets. To showcase AnySat’s flexibility, we also consider 6 datasets not included in GeoPlex. AnySat can be directly fine-tuned or linearly probed on new datasets, even if their modality combination is not featured in GeoPlex. We consider the following datasets:

- **SICKLE** [44]: A multimodal crop mapping dataset in India featuring Sentinel-1, Sentinel-2, and Landsat-8 time series. As the test set has not been released, we use the validation set.
- **BraDD-S1TS** [30]: A change detection dataset comprising Sentinel-1 time series of the Amazon rainforest, aiming to segment deforested areas.
- **TimeSen2Crop** [53]: A crop mapping dataset in Slovenia consisting of *single-pixel* Sentinel-2 time series, a modality not present in GeoPlex.
- **Sen11Flood1** [9]: A global flood mapping dataset with pixel annotations and single-date Sentinel-1 and 2 observations, a configuration not present in GeoPlex. Each tile covers 2600 hectares.
- **So2Sat** [57]: A local climate zone classification dataset containing co-registered Sentinel-1 and Sentinel-2 imagery across multiple cities worldwide, with single-date observations—a configuration not present in GeoPlex.
- **HLS Burn Scar** [41]: A dataset for burn scar detection using Harmonized Landsat-Sentinel (HLS) imagery, featuring time series data to identify post-fire affected areas and large tiles of 24K hectares.

Evaluation. We evaluate our model on the annotated datasets of GeoPlex (excluding S2NAIP-URBAN) and the 6 external datasets across three tasks: (i) **Classification:** TSAIT-TS, PASTIS-HD, PLANTED, TimeSen2Crop, So2Sat; (ii) **Semantic Segmentation:** PASTIS-HD, FLAIR, SICKLE, Sen11Flood1, HLS Burn Scars; and (iii) **Binary pixel-wise change detection:** BraDD-S1TS.

We use three evaluation settings to evaluate the models:

- **From Scratch.** The model is trained directly on the labeled training set in a supervised manner.

- **Fine-tuning.** The model is pretrained in a self-supervised manner, then fine-tuned on the training set.
- **Linear Probing.** The model is initially pretrained in a self-supervised manner, and a linear layer is fitted with the training set.

Competing Methods. We compare AnySat against state-of-the-art Earth Observation models. Most foundation models pre-trained on external data cannot be directly applied to target datasets with different input configurations. For example, the ScaleMAE and SatMAE models are trained on the Functional Map of the World [13] and limited to RGB bands, while CROMA is trained on single-date Sentinel-2 data. Since these specific modalities are not present in any of our evaluation datasets, we cannot directly evaluate these pretrained models. Instead, we modify the input layers of these models to match the target number of spectral bands.

4.2. Results and Analysis

We evaluate our model on different datasets from and outside of GeoPlex with fine-tuning and linear probing.

Performance on GeoPlex’ Test Sets. We evaluate AnySat on the test sets of the GeoPlex datasets, as shown in Fig. 5, with detailed results provided in the Appendix. Despite using a single pretrained model, AnySat sets new state-of-the-art results for TreeSatAI-TS (+0.9 weighted F1 score) and PASTIS-HD (+2.8 mIoU in classification and +0.2 in segmentation). AnySat also achieves near state-of-the-art performance on PLANTED [40], even though the ViViT models [2] were trained on a withheld dataset with nearly 80% more data of the same type. Similarly, our model performs close to the state-of-the-art on FLAIR, despite having access to only 6.5% of the extent of the Sentinel-2 tiles used by UT&T [20].

Pretraining on GeoPlex consistently improves performance, indicating that training on a collection of datasets with varied modalities leads to richer and more robust representations. The improvement is more pronounced for smaller datasets like TreeSatAI-TS and in classification tasks rather than segmentation. We attribute this to the amount of supervision available in larger datasets and dense annotations, which make pretraining less beneficial.

Performance on External Datasets. Fig. 5 shows that AnySat significantly outperforms the state-of-the-art for 6 external datasets, improving SICKLE by +3.6 mIoU, BraDD-S1TS by +10.2 mIoU, and TimeSen2Crop by +11.0 OA. These gains highlight AnySat’s strong spatial generalization as GeoPlex primarily covers the northern hemisphere, while the external datasets have global coverage.

Moreover, AnySat can be effectively linearly probed for semantic segmentation. It surpasses all specialized ap-

proaches on BraDD-S1TS when linearly probed, and likewise exceeds the performance of foundation models with fine-tuned UperNet segmentation heads on Sen1Flood11. Notably, a linearly probed AnySat outperforms a fine-tuned Prithvi2 [47] on Sen1Floods11 with 10^5 fewer free parameters. These findings underscore the expressive power of AnySat’s self-supervised features and confirm that it can be adapted to new tasks and datasets at minimal training cost and still deliver competitive performance.

Performance on New Sensor Configurations. We demonstrate AnySat’s robustness in handling sensor configurations not present in GeoPlex. For instance, SICKLE’s LandSat8 requires three additional bands beyond those used in S2NAIP’s LandSat8, while TimeSen2Crop provides only 9 of the 10 bands employed by our Sentinel-2 projector network. Applying the padding strategy described in Sec. 3.3, AnySat achieves state-of-the-art results on both datasets. We also evaluate AnySat on single-date Sentinel images (So2Sat, Sen1Flood11) and single-pixel time series (TimeSen2Crop), which were never part of GeoPlex, and again observe state-of-the-art performance. Finally, we test AnySat on the HLS-BurnScar dataset [41]. As GeoPlex does not contain HLS data (but contains Sentinel and LandSat), we train a new projector for this new modality. AnySat outperforms all competing methods, including Prithvi [29], which was trained on 252M km² of HLS imagery. In comparison, GeoPlex comprises only 249K km² without any HLS data, further illustrating the strong generalization capability of AnySat.

Ablation Study. We evaluate the impact of several key design choices and report the results in Tab. 1. All results are presented for the Fold 5 of PASTIS-HD and for the classification and semantic segmentation tasks. We do not pretrain on the entire GeoPlex but use Fold 1 to 4 of PASTIS-HD in a self-supervised fashion.

- **Random Token Dropping.** We replaced JEPA’s block masking strategy with purely random token dropping for the student network. This modification decreased classification performance but slightly improved segmentation results. In order to use a single model configuration for all tasks, we maintained a unified approach. Interestingly, block masking does not appear to be as critical for EO data than for natural images (see Table 6 in [3]).
- **No Contrastive Loss.** We remove the contrastive loss and retain only the reconstruction loss $\mathcal{L}_{\text{JEPA}}$. This substantially reduces the classification performance (-4.3 F1) but only a moderate decrease in segmentation performance (-0.2 mIoU). These findings suggest that the contrastive loss can help the feature-predictive approach learn more discriminative features, particularly benefiting classification tasks.

Table 1. **Ablation.** We evaluate the impact for several critical design choices of our model on the Fold 1 of PASTIS-HD.

Experiment	classification macro F1	segmentation mIoU
best configuration	72.0	63.6
random token dropping	71.3	64.1
no contrastive	67.7	63.4
naive semseg	-	61.2

- **Naive Semantic Segmentation.** We predict pixel-wise logits directly from the patch embeddings without utilizing subpatch features. This results in a decrease in segmentation performance by 2.4 mIoU, highlighting the importance of subpatches in providing fine-grained spatial information.

Inference and Training Times. Our model was pretrained on GeoPlex using 1760 GPU-hours on an NVIDIA H100 GPU. Fine-tuning takes between 10 and 40 hours, depending on the dataset size. Linear probing takes approximately 2 hours on BraDD-S1TS.

In terms of inference speed, AnySat processes one monochrome tile from TreeSatAI [1] in 3ms on average, which is faster than ScaleMAE [42] (10ms) and comparable to DOFA [54] (3ms) and OmniSat [4] (2ms).

5. Conclusion

We have presented AnySat, a versatile architecture designed to address the diversity of EO data in terms of resolutions, scales, and modalities. By leveraging a joint embedding predictive architecture and scale-adaptive spatial encoders, AnySat can be trained in a self-supervised manner on highly heterogeneous datasets. Pretrained on GeoPlex, a comprehensive collection of multimodal datasets with varying characteristics, our model achieved state-of-the-art performance across multiple datasets, tasks, and modalities.

A key advantage of AnySat is its ability to be applied and fine-tuned on a wide array of combinations of data types and scales with a single model. Moreover, new datasets can be easily incorporated into GeoPlex for self-supervised pre-training. Our goal is to generalize this approach to develop a versatile foundation model for environmental monitoring on a global scale.

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